Lecture 12: Slutsky, and testing

1 Where are we?

- Last time, we wrapped up the relationship between solutions to the consumer problem,
  the saddle points of the Lagrangian,
  and solutions to the Kuhn-Tucker condition
- We did a quick example of solving the consumer problem with Lagrangians,
  and saw how the Lagrangian gave an easy Envelope Theorem proof of Roy's Identity
- And we introduced the Expenditure Minimization Problem and Hicksian demand,
  and proved some properties of $h(p, u)$ and $e(p, u)$
- Today, some more properties of $h(p, u)$ and $e(p, u)$,
  the link between Marshallian and Hicksian demand,
  and how we can use Hicksian demand to test our model

- First... any questions?

2 Back to Expenditure Minimization

- We defined the Expenditure Minimization Problem as

$$\min_{x \geq 0} p \cdot x \quad \text{subject to} \quad u(x) \geq u$$

with

$$e(p, u) \quad = \quad \min_{x \geq 0} p \cdot x \quad \text{subject to} \quad u(x) \geq u$$

the value function, which we called the expenditure function, and

$$h(p, u) \quad = \quad \arg\min_{x \geq 0} p \cdot x \quad \text{subject to} \quad u(x) \geq u$$

the solution, which we called Hicksian demand (or compensated demand)

- We showed if u is continuous, prices are strictly positive, and there's some x that achieves the target utility level,

  then the expenditure minimization problem has a solution,

  or Hicksian demand is non-empty

- We proved some basic properties of h and e,

  most of which were analogues of properties we already saw in producer theory –

  probably the most interesting were the No Excess Demand property,

  that if u is continuous, Hicksian demand gives utility exactly u and no more;

  and that the expenditure function e is concave in prices

- So, on to some more properties

- **Proposition (Shepard's Lemma).** Let u be a continuous utility function representing LNS preferences, and suppose $h(p,u)$ is single-valued. Then $e(p,u)$ is differentiable in p, and

$$\frac{\partial e}{\partial p_i}(p,u) \quad = \quad h_i(p,u)$$

- Proof is via the Envelope Theorem:

$$e(p,u) \quad = \quad \min_{\{x:u(x)\geq u\}} \{p_1x_1 + \ldots + p_kx_k\}$$

so

$$\frac{\partial}{\partial p_i}e(p,u) \quad = \quad \frac{\partial}{\partial p_i}\left(p_1x_1 + \ldots + p_kx_k\right)|_{x=h(p,u)} \quad = \quad x_i|_{x=h(p,u)} \quad = \quad h_i(p,u)$$

- **Proposition (Law of Demand).** Suppose $p, p' \geq 0$, $x \in h(p,u)$, and $x' \in h(p',u)$. Then

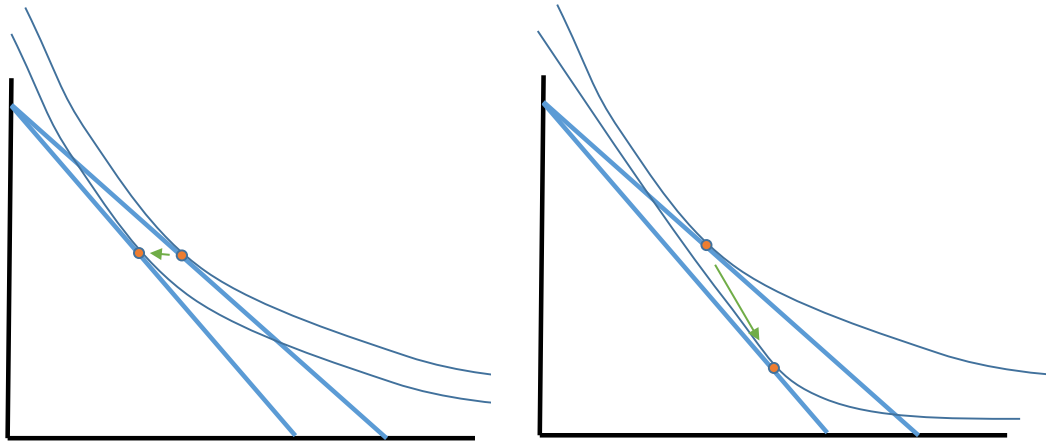
$$(p'-p)(x'-x) \quad \leq \quad 0$$

- Like with the Law of Supply, $p'-p$ is a vector pointing in the direction of a price change;
  $x'-x$ is a vector pointing in the direction of the change in Hicksian demand;
  and the negative dot product means these go in opposite directions
- We'll skip the proof, since it's analogous to the Law of Supply
- And of course, if p' and p differ in just one price, we get

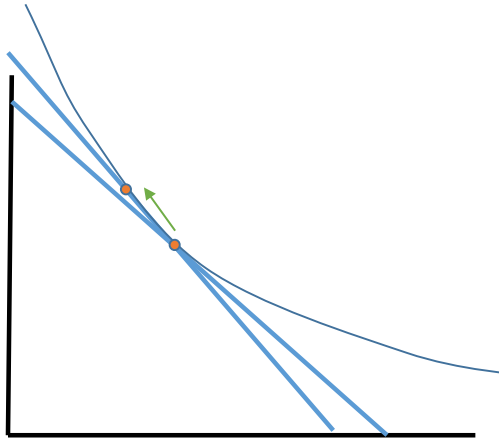
$$(p_i'-p_i)(h_i(p',u)-h_i(p,u)) \leq 0$$

  or, if $p_i' > p_i$, $h_i(p',u) \leq h_i(p,u)$
- Or, the Hicksian demand for a good is decreasing in that good's price
- While this seems obvious, this is *not* always true with Marshallian demand

- With Marshallian demand, when p_1 goes up, you shift to a different (lower) indifference curve; since the utility function could have any shape, the effect on x_1 is ambiguous



- But with Hicksian demand, when p_1 goes up, since utility stays constant, you move along the same indifference curve – so the effect on h_1 is clear



- (In fact, there's a name for goods where the Marshallian demand is increasing in its own price, we'll get to that soon)

3 More results when h is single-valued and differentiable

- If Hicksian demand is single-valued and differentiable,

  consider the matrix of its partial derivatives with respect to price:

$$D_p h(p,u) \quad = \quad \begin{bmatrix} \frac{\partial h_1}{\partial p_1} & \frac{\partial h_1}{\partial p_2} & \cdots & \frac{\partial h_1}{\partial p_k} \\ \\ \frac{\partial h_2}{\partial p_1} & \frac{\partial h_2}{\partial p_2} & \cdots & \frac{\partial h_2}{\partial p_k} \\ \\ \vdots & \vdots & & \vdots \\ \\ \frac{\partial h_k}{\partial p_1} & \frac{\partial h_k}{\partial p_2} & \cdots & \frac{\partial h_k}{\partial p_k} \end{bmatrix}$$

- Here are some properties that should look pretty familiar from producer theory:

Proposition. Suppose u represents a preference relation $\succsim$,
and $h(p,u)$ is single-valued and differentiable at (p,u), with $p \gg 0$.
Then the matrix $D_p h(p,u)$ is symmetric and negative semi-definite, and $[D_p h(p,u)]p = 0$.

- The first two come straight from Shepard's Lemma –

  since $h_i = \frac{\partial e}{\partial p_i}$, this is the matrix of second derivatives of e,

  which is symmetric (since cross-partials are symmetric)

  and negative semidefinite (since e is concave)

- But stepping back from the math, the symmetry result is pretty surprising –

  the change in compensated demand for butter due to a change in price of beer,

  is the same as the change in demand for beer with a change in the price of butter

- In fact, it's a strong prediction we could potentially test with data,

  if we could observe Hicksian demand

  (foreshadowing)

- Negative semidefinite also implies the diagonal elements are negative – $\frac{\partial h_i}{\partial p_i} \leq 0$ –

  which we already knew from the Law of Demand

- For the last result, $[D_p h]p = 0$, the textbook offers a couple of mechanical ways to prove this, but there's a more (I think) intuitive explanation
- Knowing $D_p h$ is symmetric, we can use its transpose;
  matrix math gives

$$(D_p h(p,u))^T p \quad = \quad \begin{bmatrix} \frac{\partial h_1}{\partial p_1} & \frac{\partial h_2}{\partial p_1} & \cdots & \frac{\partial h_k}{\partial p_1} \\ \frac{\partial h_1}{\partial p_2} & \frac{\partial h_2}{\partial p_2} & \cdots & \frac{\partial h_k}{\partial p_2} \\ \vdots & \vdots & & \vdots \\ \frac{\partial h_1}{\partial p_k} & \frac{\partial h_2}{\partial p_k} & \cdots & \frac{\partial h_k}{\partial p_k} \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ \vdots \\ p_k \end{bmatrix} \quad = \quad \begin{bmatrix} \frac{dh}{dp_1} \cdot p \\ \frac{dh}{dp_2} \cdot p \\ \vdots \\ \frac{dh}{dp_k} \cdot p \end{bmatrix}$$

  so saying this is 0 is the same as saying that

$$\frac{dh}{dp_i} \cdot p \quad = \quad 0$$

  for each p_i
- We know that Hicksian demand always achieves the target utility level u,
  so as p_i changes, the change in Hicksian demand is movement along an indifference curve
- $\frac{dh}{dp_i} \cdot p = 0$ is saying the indifference curves are perpendicular to the price vector –
  which is the same as saying the indifference curve is tangent to the budget line –
  which we know must be true at the optimum!

4 Relating Marshallian and Hicksian Demand

- So, Marshallian Demand is the solution to utility maximization,

$$\max u(x) \quad \text{subject to} \quad p \cdot x \leq w$$

and Hicksian demand is the solution to expenditure minimization,

$$\min p \cdot x \quad \text{subject to} \quad u(x) \geq u$$

- As I've hinted at before,
  I think of Marshallian demand as more or less the problem people face in the real world;
  but Hicksian demand is something we have a lot of strong theoretical results about
- Next, we'll see how the two relate to each other,
  and in particular, how we can infer Hicksian demand from Marshallian demand;
  so if we observe someone's Marshallian demand,
  we can test whether their Hicksian demand is consistent with the properties we just saw,
  to test whether their choices are consistent with utility maximization

Proposition. Suppose u is a continuous utility function representing LNS preferences on $\mathbb{R}^k_+$. Suppose $p \gg 0$, $w \geq 0$, and $u \geq u(0)$.

1. Fixing p, "e and v are inverses":

$$e(p, v(p, w)) \quad = \quad w \qquad \text{and} \qquad v(p, e(p, u)) \quad = \quad u$$

2. Hicksian and Marshallian demand match at corresponding wealth/utility levels:

$$h(p, u) \quad = \quad x(p, e(p, u)) \qquad \text{and} \qquad x(p, w) \quad = \quad h(p, v(p, w))$$

- Proofs.
  - Fix p and w, and pick $\hat{x} \in x(p, w)$
  - Since preferences are LNS, $p \cdot \hat{x} = w$; and define $u = v(p, w) = u(\hat{x})$
  - Since $\hat{x}$ gives us one way to achieve utility u at cost w, $e(p, u) \leq w$
  - But if there were a way to achieve u at cost less than w, this would give another bundle $x' \in x(p, w)$ with $p \cdot x' < w$, which would violate Walras' Law, so $e(p, u) \geq w$
  - Thus, $e(p, u) = w$ when $u = v(p, w)$, or $e(p, v(p, w)) = w$

  - Next, let's do $v(p, e(p, u)) = u$
  - Pick $\hat{x} \in h(p, u)$, so that $e(p, u) = p \cdot \hat{x}$
  - By no-excess-utility, $u(\hat{x}) = u$
  - Of course, this means if we let $w = e(p, u)$, we must have $v(p, w) \geq u$, since one option is to consume $\hat{x}$
  - On the other hand, if $v(p, w)$ were strictly greater than u, this would mean there was some other bundle x' with $p \cdot x' \leq w = p \cdot x$ giving $u(x') > u$
  - Since $\hat{x} \in h(p, u)$ and x' is the same cost, $x' \in h(p, u)$ as well; but this would violate the no-excess-utility condition

  - Finally, to show equivalence of Marshallian and Hicksian demand,
    we can write Marshallian demand as

$$x(p, w) \quad = \quad \{x : u(x) \geq v(p, w)\} \cap \{x : p \cdot x \leq w\}$$

    and Hicksian demand as

$$h(p, u) \quad = \quad \{x : u(x) \geq u\} \cap \{x : p \cdot x \leq e(p, u)\}$$

  - If we plug in $u = v(p, w)$ and use the fact that $e(p, v(p, w)) = w$, these are the same, so $x(p, w) = h(p, v(p, w))$
  - Similarly, if we plug in $w = e(p, u)$ and use the fact that $v(p, e(p, u)) = u$, they're the same, so $x(p, e(p, u)) = h(p, u)$

- So we've shown "equivalence" between the solutions to the two problems
- We can think of $e(p, v(p, w)) = w$ as basically a restatement of Walras' Law:

  whatever maximal utility $v(p, w)$ you can get for wealth w,

  the cheapest way to get it costs the full w
- And similarly, $v(p, e(p, u)) = u$ is basically No Excess Utility:

  however much it costs you to buy utility u, you can't get higher utility for the same cost
- And equivalence of the solutions to the two problems is bascially the fact that in either case, the solution can be thought of as the intersection of two sets –

  the set of bundles weakly better than x, and the set of bundles weakly cheaper than x

- However, saying that when $u = v(p, w)$, Hicksian and Marshallian demand coincide does not mean they are always identical
- The whole point is that they respond differently to price changes
- That is, when a price changes, Hicksian and Marshallian demand respond differently –

  because Hicksian demand must leave utility unchanged,

  while Marshallian demand must leave budget unchanged
- How the two relate in terms of changes brings us to the Slutsky Equation

5 The Slutsky Equation

- Begin with

$$x(p, e(p, u)) \quad = \quad h(p, u)$$

which we just proved

- Focus on good i,

$$x_i(p, e(p, u)) \quad = \quad h_i(p, u)$$

- Differentiate with respect to the price of good j,

$$\frac{\partial x_i}{\partial p_j}(p, e(p, u)) + \frac{\partial x_i}{\partial w}(p, e(p, u))\frac{\partial e}{\partial p_j}(p, u) \quad = \quad \frac{\partial h_i}{\partial p_j}(p, u)$$

- Recall that $\frac{\partial e}{\partial p_j} = h_j$ (Shepard's Lemma), and rearrange, giving

$$\frac{\partial x_i}{\partial p_j}(p, e(p, u)) \quad = \quad \frac{\partial h_i}{\partial p_j}(p, u) - \frac{\partial x_i}{\partial w}(p, e(p, u))h_j(p, u)$$

- Letting $u = v(p, w)$, so $e(p, u) = w$, and noting $h_j(p, u) = x_j(p, w)$, this is

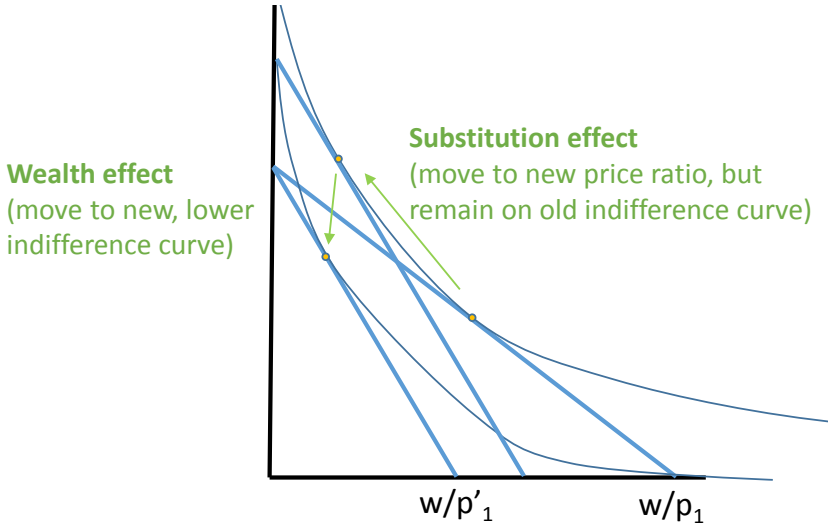
$$\frac{\partial x_i}{\partial p_j}(p, w) \quad = \quad \frac{\partial h_i}{\partial p_j}(p, u) - \frac{\partial x_i}{\partial w}(p, w)x_j(p, w)$$

which is called the Slutsky Equation

- **Theorem. (Slutsky Equation.)** Let u be a continuous utility function representing LNS preferences on $\mathbb{R}^k_+$; pick $p \gg 0$, and let $w = e(p, u)$. If $h(p, u)$ and $x(p, w)$ are single-valued and differentiable at (p, u, w), then

$$\frac{\partial x_i}{\partial p_j}(p, w) \quad = \quad \frac{\partial h_i}{\partial p_j}(p, u) \quad - \quad \frac{\partial x_i}{\partial w}(p, w)x_j(p, w)$$

- This lets us decompose $\frac{\partial x_i}{\partial p_j}$ into two effects – a **substitution effect** and a **wealth effect**
  - From a Marshallian perspective, an increase in p_j does two things –
    it makes good j *relatively* more expensive, and it makes you effectively poorer
  - That is, it changes the cheapest way to achieve a particular level of utility,
    and it reduces the level of utility you expect to achieve
  - The first effect is how Hicksian demand responds
  - The second effect is how Marshallian demand responds to wealth, times how much your wealth goes down
  - How much does your effective wealth go down?
    Well, if p_j goes up by ϵ, then for every unit of good j that you were planning to consume, you're now ϵ over your budget constraint –
    so you're effectively made poorer by ϵx_j for every ϵ the price rises
  - So the effect is whatever the effect of losing some wealth is on your Marshallian demand, multiplied by x_j



- Note that both the substitution and wealth effects can have either sign,
  depending on the nature of the goods
- The Slutsky equation still holds when $i = j$, giving

$$\frac{\partial x_i}{\partial p_i}(p, w) \quad = \quad \frac{\partial h_i}{\partial p_i}(p, u) \quad - \quad \frac{\partial x_i}{\partial w}(p, w) x_i(p, w)$$

  where here, we know the substitution effect is negative ($\partial h_i / \partial p_i \leq 0$),
  but the wealth effect can still have either sign

6 Inferior Goods, Giffen Goods, Substitutes, Complements

- In fact, we have names for the different signs these effects can have...

- Good i is a **normal good** if x_i is increasing in w,

  and an **inferior good** if x_i is decreasing in w

  Luxury goods tend to be normal – when you get richer, you consume more BMWs and caviar than you used to.

  Inferior goods might be things like canned tuna or ramen – you consume them a lot when you're poor, but then when you get richer, you start swtiching to other, "better" goods, so your demand decresaes

- Good i is a **regular good** if x_i is decreasing in p_i,

  and a **Giffen good** if x_i is increasing in p_i

  Going back to the Slutsky equation, a Giffen good is an inferior good where the wealth effect overwhelms the substitution effect

  The "classic" example of a Giffen good is potatoes during the Irish potato famine. The poor subsisted largely on potatoes, but had some money left over for other things (like other foods). When the price of potatoes went up, they still needed to eat enough to survive, and potatoes were still the cheapest option, so people ate *more* potatoes because it's all they could afford.

  There's a 2008 experimental paper that finds this effect for rice and wheat in two provinces in China.[1]

  Another weird recent sort-of-example was US government debt. In August 2011, the U.S. government was in a political fight over whether to increase the debt ceiling, and due to the uncertainty, Standard and Poors downgraded US debt from AAA (their highest rating) to AA (their second-highest). This led to an increase in demand for US debt, driving prices up rather than down. Why? Even if it was being judged to be riskier than before, US debt is still perceived as the safest investment available; the added riskiness led to a "flight to quality" and attracted more investment in the still-pretty-safe asset![2]

[1] Jensen and Miller (2008), "Giffen Behavior and Subsistence Consumption," *American Economic Review* 98.4

[2] G. I., "Treasury Potatoes – Bonds as Giffen Goods," *The Economist* Nov 22 2011; also see Komal Sri-Kumar, "The Bond Market Is Now a Giffen Good," *Bloomberg*, Jul 12 2019, for a more recent example of the same phenomenon.

- Good i is a **substitute** for good j if h_i is increasing in p_j,
  and a **complement** if h_i is decreasing in p_j
- Note that this is symmetric:
  if i is a substitute for j, then j is a substitute for i, since $\partial h_i/\partial p_j = \partial h_j/\partial p_i$
- But two goods could be substitutes at some prices and utility levels and complements at others
  (If you're incredibly poor, maybe peanut butter and jelly are substitutes,
  because you're just trying to get enough calories to survive and you'll eat whichever is cheaper;
  but at higher income levels, they're complements,
  because you choose to eat them together as opposed to other alternatives like ham and cheese.)
- Also note that if there are only two goods, they have to be substitutes –
  since h_i decreases in p_i, you need to consume more of j to get the same level of utility.
  We need 3 or more goods to have complements in the Hicksian sense.

- Finally, good i is a **gross substitute** for j if x_i is increasing in p_j,
  and a **gross complement** if x_i is decreasing in p_j
- This one is *not* always symmetric,
  because the wealth effects of the two goods may be different
- If we say two goods are gross substitutes, we mean each is a gross substitute for the other.
- And again, two goods could be substitutes in one price range but complements in another.

7 Using Slutsky to Test our Theory

- Recall the conclusion we reached earlier – the matrix of partial derivatives of Hicksian demand with respect to prices is symmetric and negative semi-definite
- That would be a nice conclusion to test, if we could observe someone's Hicksian demand
- While we don't expect to actually observe Hicksian demand,

  the Slutsky equation lets us infer its partial derivatives from something we can observe,

  Marshallian demand
- Rewrite the Slutsky equation as

$$\frac{\partial x_i}{\partial p_j}(p,w) \quad + \quad \frac{\partial x_i}{\partial w}(p,w)x_j(p,w) \quad = \quad \frac{\partial h_i}{\partial p_j}(p,u)$$

- What's nice about this is:
  - The right-hand side is something we have very strong theoretical predictions about
  - The left-hand side is something we could potentially observe
- That is, if we can observe a consumer's Marshallian demand, we can construct the matrix

$$D_p h(p,u) \quad = \quad \begin{bmatrix} \frac{\partial x_1}{\partial p_1} + \frac{\partial x_1}{\partial w}x_1 & \cdots & \frac{\partial x_1}{\partial p_k} + \frac{\partial x_1}{\partial w}x_k \\ \vdots & & \vdots \\ \frac{\partial x_k}{\partial p_1} + \frac{\partial x_k}{\partial w}x_1 & \cdots & \frac{\partial x_k}{\partial p_k} + \frac{\partial x_k}{\partial w}x_k \end{bmatrix}$$

- And we know that to be consistent with our theory, this matrix needs to be symmetric and negative semidefinite
- In fact, along with other easy-to-check conditions, that's also sufficient:
- **Theorem.** A differentiable demand function $x(p,w)$ is consistent with utility maximization given rational, locally non-satiated preferences if and only if:
  1. $x(p,w)$ is homogeneous of degree 0
  2. At every (p,w), $p \cdot x(p,w) = w$ (Walras' Law)
  3. At every (p,w), the matrix $[\frac{\partial x_i}{\partial p_j} + \frac{\partial x_i}{\partial w}x_j]$ is symmetric and negative semidefinite
- To put it another way, our theory of rational choice predicts these properties –

  and nothing else!

- So this is nice – if we observe someone's entire demand function $x(p, w)$ at all prices and wealth levels, we can determine whether they're rational utility maximizers or not –

  by testing whether the Slutsky matrix coming from x is symmetric and negative semidefinite

- There are some empirical papers that do exactly that

- I don't know the empirical literature well, but I get the sense the early literature focused on symmetry of the Slutsky matrix,

  and tended to rely on estimating a parametric demand system and seeing if the parameters implied Slutsky-symmetry;

  some more recent work has included testing for negative semidefiniteness,

  and been more nonparametric

- I can at least point to you some relatively recent papers by authors I would trust:

  Arther Lewbel (1995), "Consistent Nonparametric Hypothesis Tests with an Application to Slutsky Symmetry," *Journal of Econometrics* 67

  B. Haag, S. Hoderlein and K. Pendakur (2009), "Testing and Imposing Slutsky Symmetry in Nonparametric Demand Systems," *Journal of Econometrics* 153 – focus on proper econometric testing

  Stefan Hoderlein (2011), "How Many Consumers Are Rational?" *Journal of Econometrics* 164 – focus on allowing for unobserved heterogeneity across consumers when using aggregate data

  Victor Aguiar and Roberto Serrano (2017), "Slutsky Matrix Norms: The Size, Classification, and Comparative Statics of Bounded Rationality," working paper – focus is to measure distance between observed Slutsky matrix and nearest matrix satisfying required conditions, and decompose violations into "inattentiveness to changes in purchasing power," "money illustion" (nominal rather than real prices), and actual violations of the Law of Demand

- So that's rationalizability:

  when are observations consistent with our model of rational choice?

  when they imply a Slutsky matrix which is symmetric and negative semidefinite

- (That's for the case of "infinite data" – when we observe the entire demand function;

  next class, we'll consider the question of finite observations)

- We can also think about whether we can recover preferences from observation of demand,

  or whether our model is identified...

8 Recovering Preferences from $x(p, w)$

- Suppose we know Marshallian demand $x(p, w)$ at every (p, w)

- First of all, recall that Marshallian demand is determined by preferences, not utility –

  that is, the solution to the consumer problem is the same for any utility function representing the same preferences

- So if we start with demand, the best we can hope to do is recover ordinal preferences,

  not a unique cardinal utility function

- Which leaves us with a degree of freedom – we can normalize cardinal utility however we want, since we're going to recover the same preferences however we do it

- So let's fix a reference price vector p^0,

  and normalize utility such that $v(p^0, w) = w$, or $e(p^0, u) = u$

- (That is, since there's no natural scale for utility,

  we can pick a reference price vector,

  and just define the cardinal level of utility as the amount of money it takes to afford it at prices p_0)

- Now pick a value for u; for each good i, we know that

$$\frac{\partial e}{\partial p_i}(p, u) \quad = \quad h_i(p, u) \quad = \quad x_i(p, e(p, u))$$

- Since $x_i(\cdot, \cdot)$ is a known function (it's observed), this is a differential equation where the unknown variable is e

- And since we can do this for each good i, we have a system of k differential equations in e, plus an initial condition $e(p^0, u) = u$

- It turns out that symmetry of the Slutsky matrix is exactly the conditions that this system has a solution (also known as "integrability" conditions);

  and negative semi-definiteness guarantees that the solution e is concave (which we know it needs to be)

- So there's some concave expenditure function e, defined for all p and u,

  that's consistent with our observed Marshallian demand

- Once we know the expenditure function $e(p, u)$, we can recover preferences from that
- We do this just like with the outer bound in producer theory,

  by defining the upper contour sets

$$V_u \quad = \quad \{x \quad : \quad u(x) \geq u\} \quad = \quad \{x \quad : \quad p \cdot x \geq e(p, u) \quad \text{for all} \quad p \gg 0\}$$

- That is, we define $u(\cdot)$ such that $u(x) \geq u$ if and only if

$$p \cdot x \quad \geq \quad e(p, u) \qquad \text{for all} \qquad p \gg 0$$

- That is, we say a consumption bundle x gives utility above u,

  if x is never strictly cheaper than the cheapest way to achieve utility u
- I'm not going to give a formal proof that this works, but the logic is this:
  - if $p^* \cdot x < e(p^*, u)$ for some p^*, then $u(x)$ has to be less than u

    because if $u(x)$ were at least u, then x would give a way to achieve target utility u,

    and therefore $e(p^*, u)$ couldn't be more than the cost of x
  - if x is chosen at some p^*, then $p^* \dot{x} = e(p^*, u)$
  - so for any x, if $u(x) \geq u$,

    we need $p \cdot x \geq e(p, u)$ for every p,

    with equality holding for some p if x is sometimes chosen
  - For bundles x which are never chosen, it's not obvious this gives the "correct" preferences, but it does turn out to work
  - (The proof is in MWG p 77)

- Since we're inferring preferences from choices, there's no guarantee we match the original preferences on bundles that never get chosen;

  all we're guaranteed is that we've found preferences that generate the same Marshallian demand as we started with.

9 One last point – rational choice

- While we're in the midst of testing when consumer behavior is consistent with rational choice...
  it seems worth saying at least a word about the alternative
- And you all know, there's a large and growing literature on behavioral economics,
  showing that people don't always act in accordanace with rational choice
- We're not going to touch on that in here
- In some sense, we don't "need" people to *always* act rational,
  to believe there's value in understanding a model of rational choice
- David Friedman – son of Milton Friedman,
  and a serious libertarian economist and excellent writer – put it well:[3]
  "One summer, a colleague asked me why I had not bought a parking permit.
  I replied that not having a convenient place to park made me more likely to ride my bike.
  He accused me of inconsistency. As a believer in rationality, I should be able to make the correct choice between sloth and exercise without first rigging the game.
  My response was that rationality is an assumption I make about other people.
  I know myself well enough to allow for the consequences of my own irrationality.
  But for the vast mass of my fellow humans, about whom I know very little, rationality is the best predictive assumption available."

- yes, in specific contexts,
  people may behave in seemingly (and even systematically) "irrational" ways;
  and if we know enough about a specific context,
  we might have another model that offers more accurate and precise predictions
- but rational choice gives us a very sparse, simple model
  that is "pretty good" over a very wide range of domains –
  and is at very least a good starting point to learn well before building out from

[3] David Friedman (1996), *Hidden Order: The Economics of Everyday Life*, pp. 4-5